

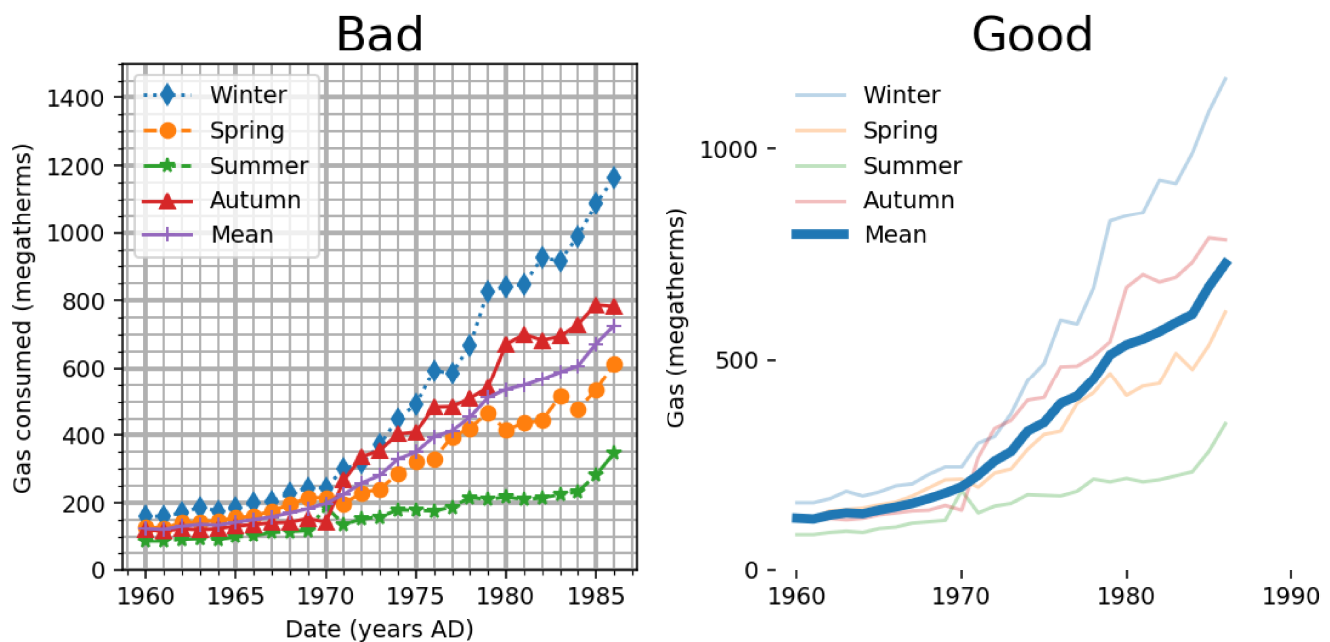
Criticising visualisations

Avoid chartjunk

Tufte proposed the principle of the **data-to-ink** ratio. As little "ink" (that is, markings) should be used to communicate as much data as possible. A simple, clean visualisation is less confusing and more aesthetically pleasing than a jumbled mess of unnecessary **chartjunk**.

Typical chartjunk includes:

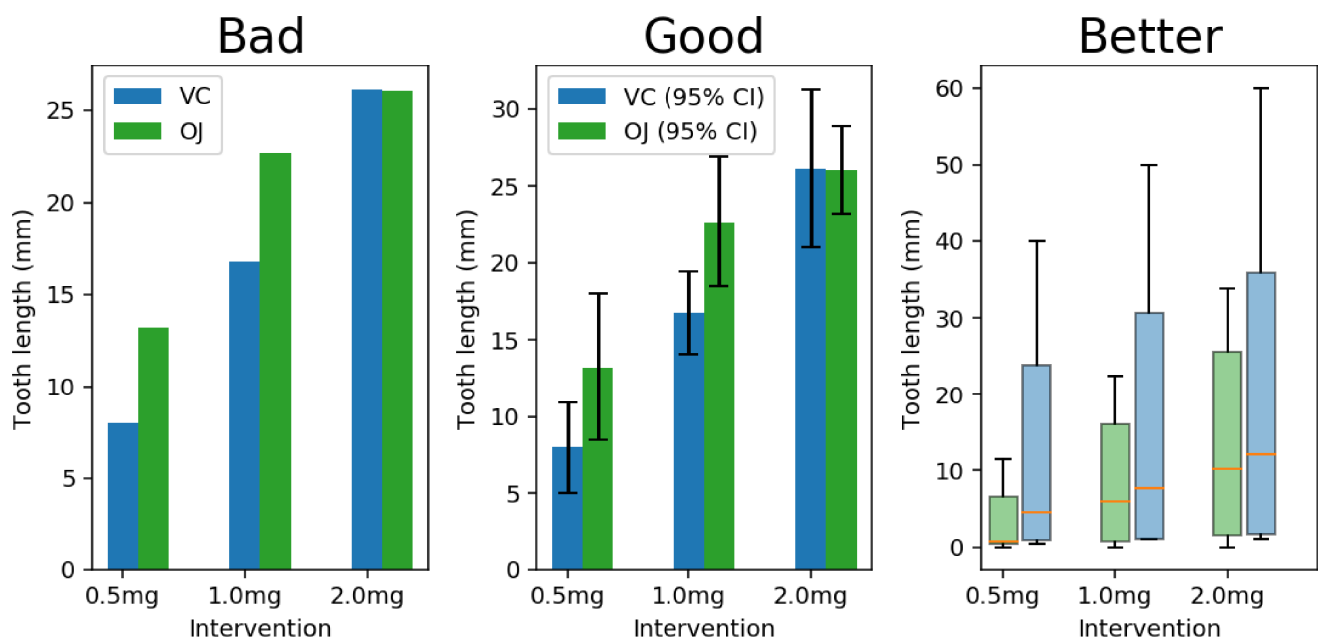
- Frames around plots
- Obtrusive grids
- Excessive marker or line styles
- Lack of visual distinction between important and unimportant elements
- Excessive tick marks
- Excessive annotations
- Clipart (just say no to clipart)



[**Left:** a dense, messy plot with lots of extraneous elements, including a dense grid, excessive tick marks and unnecessary markers. **Right** the same plot, in a clean, style with just enough elements to make the graph clear and readable.]

Represent uncertainty clearly and honestly

- Include measures of spread of a sample if these are relevant to comparing values.
- Bar charts should have error bars if they represent measures of central tendency (e.g. means, medians).
- Use Box plots or similar techniques (e.g. violin plots) if the distribution of values is useful in making judgements about differences between groups.
- Ribbon plots can be used to extend line geoms to include uncertainty.

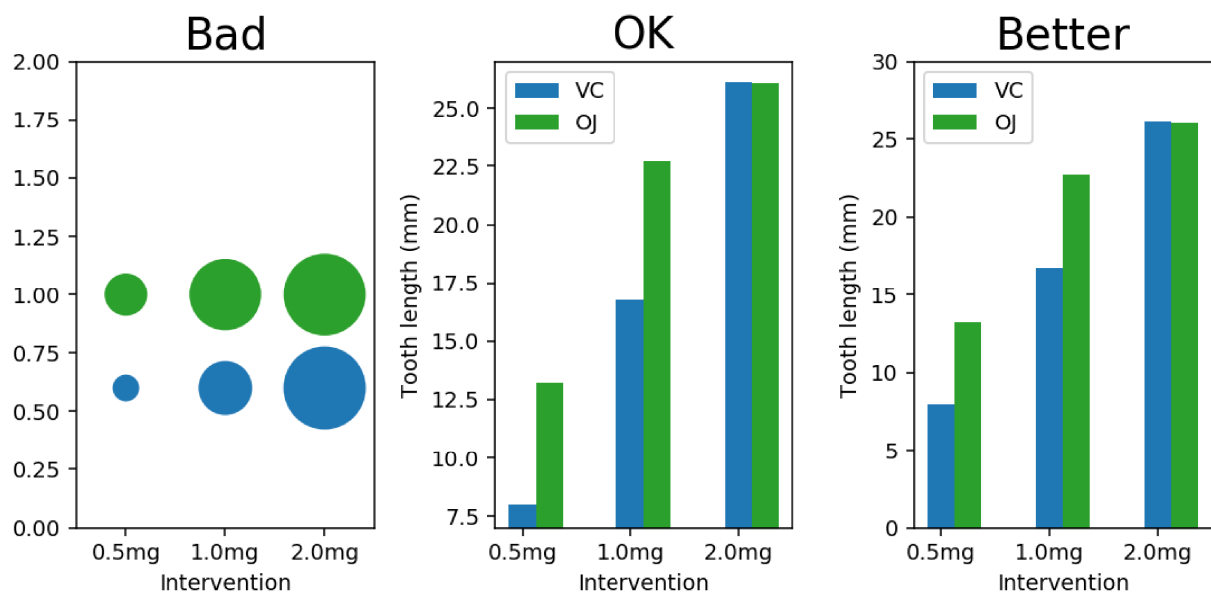


[Left blue and green represent means of two groups, but no uncertainty is shown. It is impossible to tell how big the differences between the blue and green results really are. **Centre** error bars give some indication of the spread of values. **Right** a Box plot is more complete summary and should be used whenever it is appropriate to do so]

Don't use deceptive scales

Use axis scales that inform the reader without distorting the message.

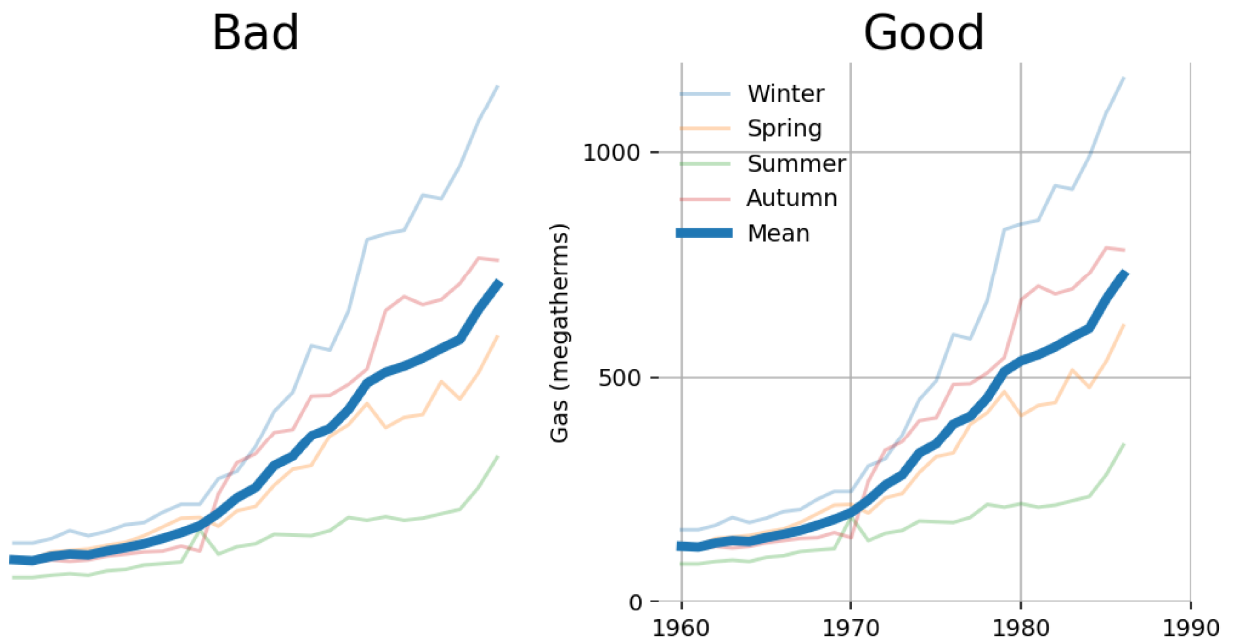
- Don't cut off data when plotting, except in extreme cases, and there you must be careful to explicitly annotate what has been cut off.
- Avoid broken axes unless absolutely necessary. Especially avoid axes that are broken somewhere other than at the origin!
- Use appropriate linear or logarithmic transformations to make the relationship clear.
- Avoid area-based (and especially volume-based) representations; you must make sure that the apparent area/volume is proportional to the value displayed and not the radius/diameter of the object and such graphs are still hard to read.



[**Left** radii of points are proportional to value; this massively inflates differences and no guide is provided to establish the units. **Centre** Visual units are now linear with data units, but the y axis scale is truncated, and the difference between the first two items is exaggerated; **Right** the y axis scale is no longer misleading]

Provide guides

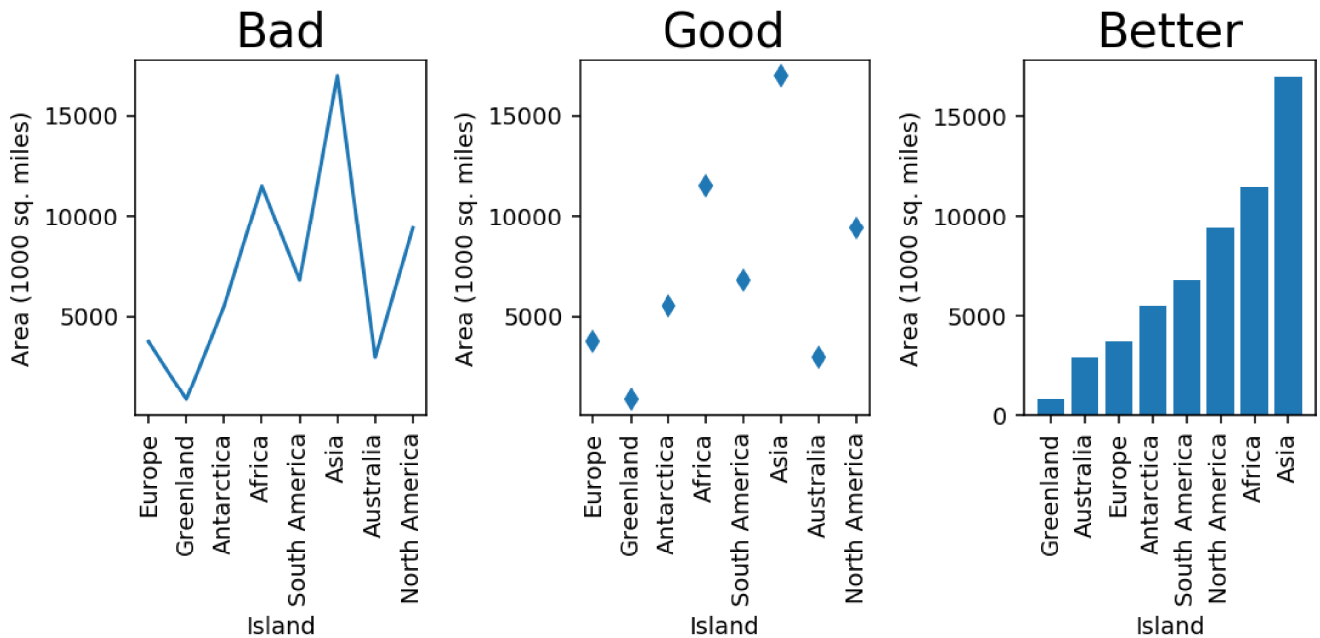
- Always provide guides for visual units (i.e. tick marks) and label axes clearly, with real-world units wherever possible.
- Always use legends to identify different layers in a plot
- Always provide a descriptive title to a plot, and a caption which explains what the figure is and what the reader should gain from looking at it.



[Left No guides at all (no axes, no grid, no legend) **Right** axis labels present, with real world units. Legend identifies layers. Grid makes it easy to look up specific values]

False connectivity

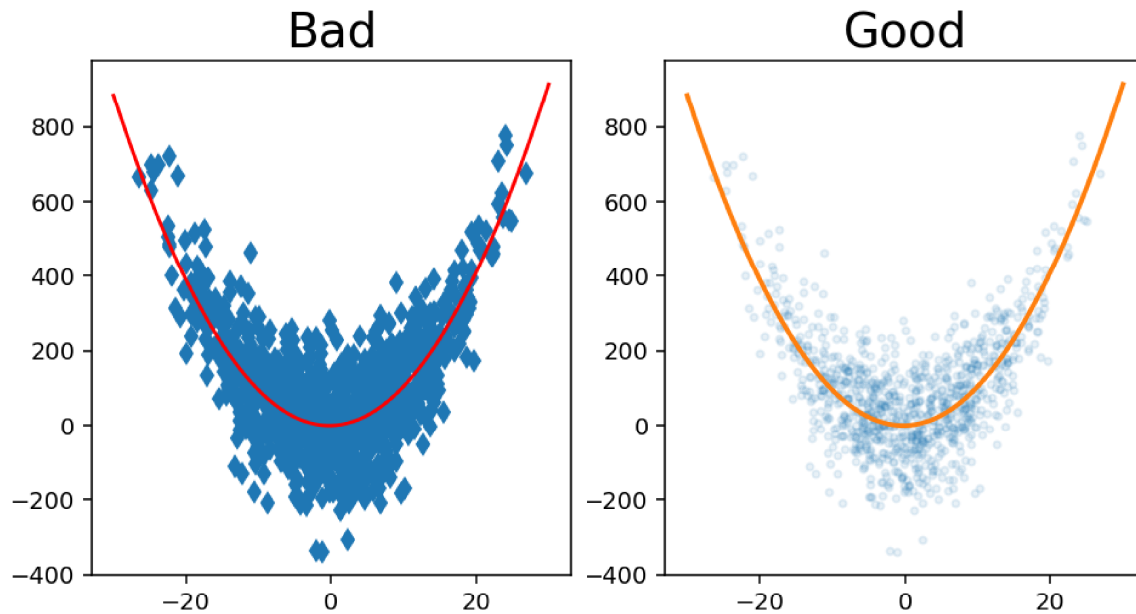
Don't connect data points with lines or curves if they don't form a continuous function. If the x axis has no defined order (categorical variable), it is even less meaningful to draw lines in between data points.



[**Left** Lines connect landmasses, but what does it mean to be in between "Greenland" and "Antartica"? The data do not form a continuous function. **Centre** No false connectivity between distinct items. **Right** A bar chart is easier to read, and by sorting the elements before plotting the graph becomes more organised and easier to read.]

Don't clutter

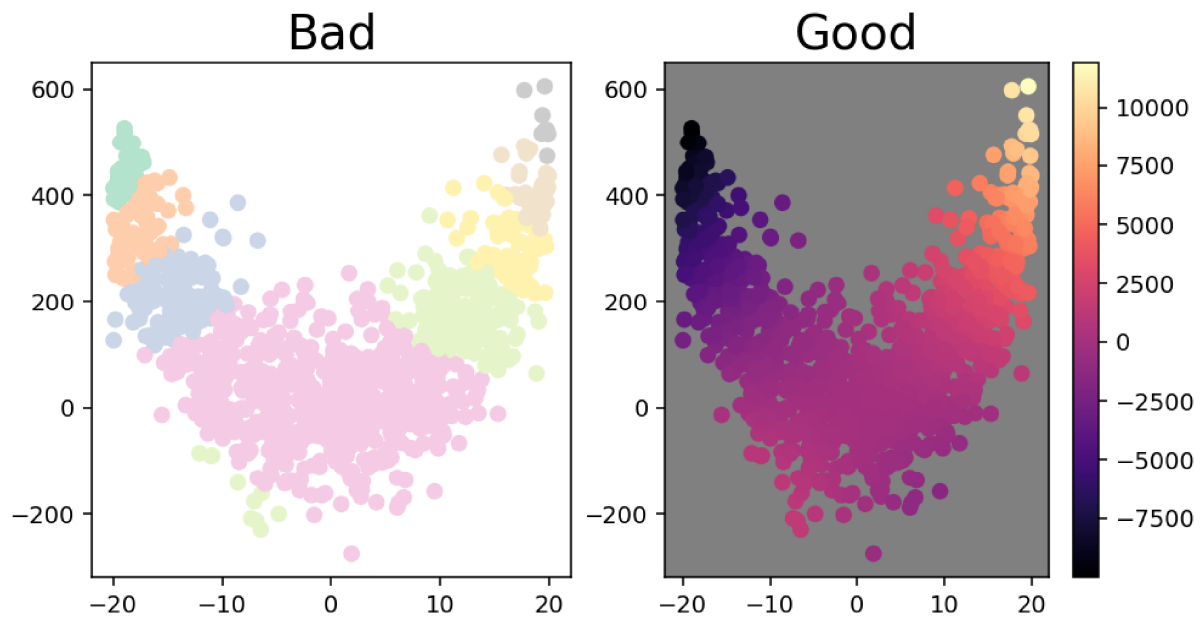
- If you have huge numbers of markers or other geoms present, consider using a lower opacity to make the plot easier to read.
- If this isn't sufficient, consider using a 2D histogram or kernel density estimate to show the **density** of points accurately (at the expense of losing the precise location of individual data points).
- Emphasise important curves or datapoints with thicker lines, larger points, but don't go overboard.



*[**Left** Extremely dense scatterplot with large markers. The points become indistinct and the whole area turns into a blue mass. **Right** Appropriate use of alpha blending (opacity) can reveal the density of points more accurately and declutter the plot.]*

Use appropriate colour scales

- If your data is unsigned (positive only), use a perceptually linear colour map with monotonically increasing brightness.
- If your data is signed (and the sign matters), use a colour map which diverges around 0.
- Scale data to colorscales appropriately, and always provide a colour bar.

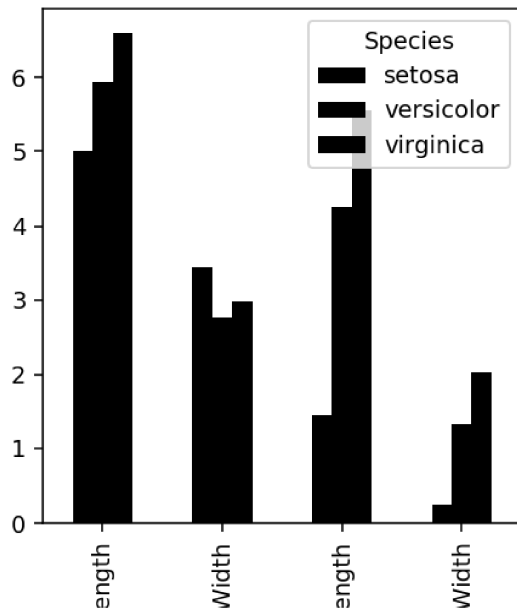


[Left The colour scale is wholly inappropriate, with discrete steps in colour introducing false contours, and a very low contrast result which is not perceptually linear in brightness. There is no guide to indicate how to interpret the colour values. **Right** Perceptually linear, monotonically increasing brightness colour map. A colour bar is provided as a guide to the true values being displayed.]

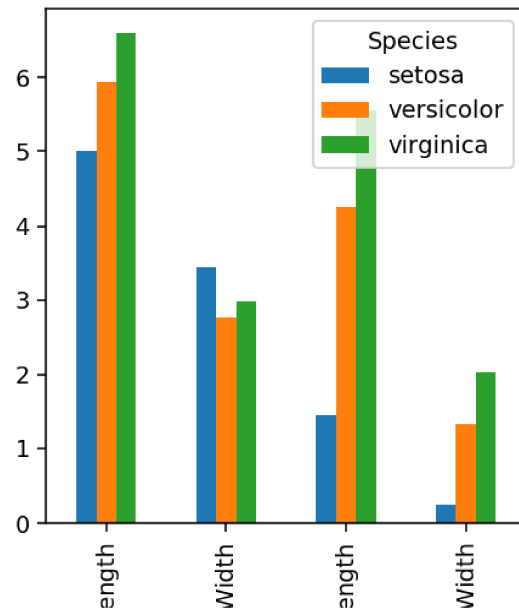
Use appropriate styling

- Make sure your geoms are distinguishable in black and white if they might appear in black and white.
- Use dashed styles, but sparingly; only 2 or 3 can easily be distinguished on a plot.
- Mark actual measurements with markers if the reader might need to know where the measurements are.

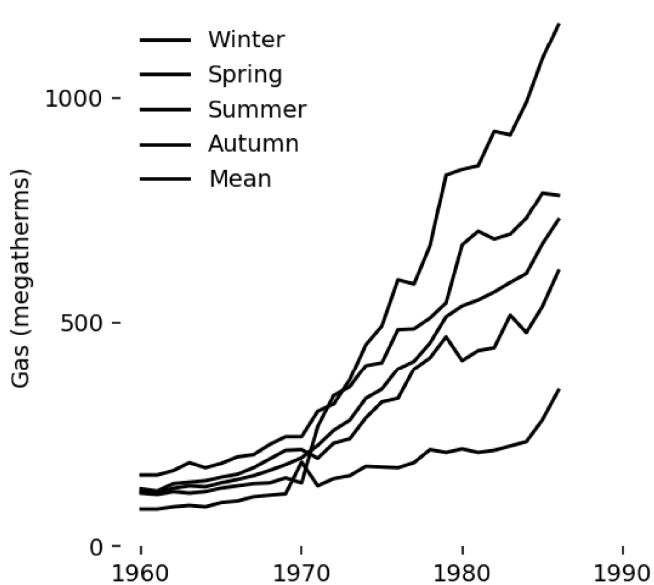
Bad



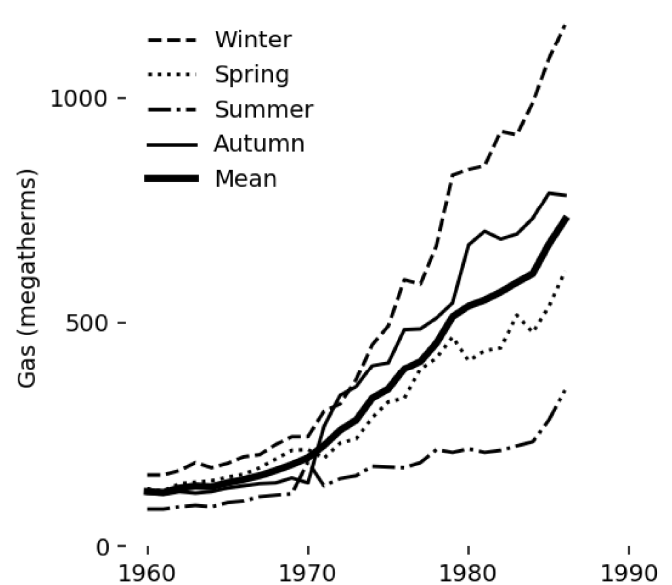
Good



Bad



Good



*[**Left** All geoms are plotted in the same colour, and same style. It is impossible to tell the different layers of the plot apart. **Right** Different colours (above) and different line styles (below) are applied to geoms to visually distinguish different layers of the plot]*